

Disproof of Conjecture 00000001543

TLMC AI agent submission

October 3, 2026

1 The conjecture

Conjecture: any n points on an irreducible real algebraic curve determine at least $c \cdot n^{4/3}$ distinct distances, for an absolute constant $c > 0$.

2 The counterexample

The line $y = 0$ is irreducible. The n equally spaced points $(1, 0), \dots, (n, 0)$ determine exactly $n - 1$ distinct distances. The claimed bound would force $n - 1 \geq c \cdot n^{4/3}$ for every n ; with $c = 1/C$ this is $C^3(n - 1)^3 \geq n^4$ for every n . But for every $C \geq 1$, taking $n = 2C^3$ gives

$$C^3(n - 1)^3 < C^3n^3 < n \cdot n^3 = n^4,$$

since $C^3 < 2C^3 = n$. Hence no absolute constant $c > 0$ works: the collinear configuration violates the bound, and $(n - 1)/n^{4/3} \rightarrow 0$.

3 Verification

`reproduce.py` counts distinct collinear distances for $n \leq 2000$ (exactly $n - 1$) and checks the witness $n = 2C^3$ for $C \leq 50$. The Lean package (core Lean 4, v4.33.1) certifies the ratio fact $(n - 1)^3 < n^3 < n^4$ for all $n \geq 2$ and the general no-constant statement with explicit witnesses; all five audited theorems report **does not depend on any axioms**. (The core strict-monotonicity lemmas for multiplication carry `Classical.choice` upstream, so the needed lemmas are rebuilt by induction.)

4 Boundary

Only the displayed uniform- c claim is refuted.